

CHE-MING, CHANG

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EDUCATION

Ph.D. student in Electrical and Computer Engineering, University of Southern California

Overall GPA: **4.0 / 4.0**

Aug., 2025 - Present

Research Topic: AI/ML-driven EDA for heterogeneous integration

B.S. in Electrical Engineering, National Taiwan University

Overall GPA: **4.17 / 4.30**

Sept., 2020 - Jun., 2024

Dean's List Award 2023 Spring

TSMC Semiconductor Advanced IC Design Program

Highlighted Coursework: Logic Synthesis and Verification, Physical Design for Nanometer ICs, Integrated Circuit Design, Computer-aided VLSI System Design, Digital Signal Processing in VLSI, Machine Learning

PUBLICATION

- [1] Pin-Han Chen, Chanhee Yang, Sungchul Joo, Hyungsoo Moon, **Che-Ming Chang**, Taek-Soo Kim, and Sung Kyu Lim, "WAFER: Warpage Analysis via Fourier-Enhanced Residual Calibration for Large-Scale 2.5D IC Designs," in *Proc. of ICCAD*, November 2026, pp. 1-9.
- [2] **Che-Ming Chang**, Prashanth Vijayaraghavan, Charles Mackin, Ashutosh Jadhav, Hsinyu Tsai, Vandana Mukherjee and Ehsan Degan, "CODMAS: A Dialectic Multi-Agent Collaborative Framework for Structured RTL Optimization," in *Proc. of EACL Industrial Track*, March 2026.
- [3] Shao-Yu Lo, **Che-Ming Chang**, Yao-Wen Chang, "Advanced Packaging Warpage Modeling with DeepONet-Based Operator Learning," in *Proc. of ICCAD*, October 2025.
- [4] **Che-Ming Chang**, Jie-Hong Roland Jiang, Dah-Wei Chiou, Ting Hsu and Guin-Dar Lin, "Quantum Circuit Compilation for Trapped-Ion Processors with the Drive-Through Architecture," in *IEEE Trans. Quantum Eng.*, vol. 6, pp. 1-14, 2025.
- [5] Cheng-Yu Chiang, Yi-Hsien Chiang, Chao-Chi Lan, Yang Hsu, **Che-Ming Chang**, Shao-Chi Huang, Sheng-Hua Wang, Yao-Wen Chang, and Hung-Ming Chen, "Mixed-Size Placement Prototyping Based on Reinforcement Learning with Semi-Concurrent Optimization," in *Proc. of ASPDAC*, January 2025, pp. 893-899.

RESEARCH EXPERIENCE

Qualcomm, Methodology team

05/2026 - 08/2026

Research Intern, Advised by Mingyu Woo

San Diego, CA

- Enhanced internal ECO-stage setup recovery and leakage power recovery flows.

Southern California Computer-Aided Design (SCCAD) Lab

Aug., 2025 - Present

Graduate Research Assistant, Advised by Prof. Sung Kyu Lim

Los Angeles, CA

- Researched in ML-based global routing optimization tasks.

Electronic Design Automation Lab

Aug., 2023 - Jan., 2025

Research Assistant, Advised by Prof. Yao-Wen Chang

Taipei, Taiwan

- Assisted in RL for concurrent macro placement project, demonstrated 12.3% / 9% additional HPWL reduction on ISPD'05 benchmarks compared to DREAMPlace / modern ML-based placer GraphPlanner.
- Developed a CNN-based operator learning framework to predict warpage in advanced packaging designs, achieving 435× speedup over 8-core finite element method while maintaining an average error of 1.9%.

IBM Almaden Research Center, AI for EDA group*Research Intern, Advised by Prashanth Vijayaraghavan*

Jun., 2024 - Sept., 2024

San Jose, CA, USA

- Developed MASCOT, a multi-agent framework for HDL code optimization by injecting and retrieving domain-specific knowledge into the code generation process, aided with compiler and simulator feedback.
- Achieved 20% / 10% improvements for larger LLMs /small fine-tuned LLMs compared to zero-shot and ReAct / Chain-of-Thought baseline on HDL optimization tasks.

Applied Logic and Computation in System Design Lab*Research Assistant, Advised by Prof. Jie-Hong Roland Jiang*

Sept., 2022 - June, 2024

Taipei, Taiwan

- Proposed an analytical method to compile a QASM circuit onto a “drive-through” modular architecture by gate partitioning followed by a fidelity-aware placement.
- Achieved 10.8% / 9.6% higher fidelity compared to modern qubit mappers SABRE / t|ket>, while ensuring the practicality and scalability of our approach.

PROJECT EXPERIENCE**Approximate Logic Synthesis with Multi-Objective Simulated Annealing***Final Project in Logic Synthesis and Verification, GitHub*

Oct., 2023 - Dec., 2023

Taipei, Taiwan

- Utilized multi-objective simulated annealing to synthesize approximate circuits within the given error bound based on the synthesis tool ABC.
- Demonstrated substantial area reduction (8-98%) over IWLS benchmarks within 1% error rate and shows quality (5-91%) and runtime improvements over the existing ALS tool ALSRAC.

QR decomposition engine for 5G MIMO demodulation*Final Project in Computer-aided VLSI System Design*

Oct., 2023 - Dec., 2023

Taipei, Taiwan

- Implemented QR decomposition engine with 2.2% soft LLR error rate in Verilog.
- Achieved 206 MHz operating frequency with 37.5 mW power under TSMC 130nm technology with full VLSI design flow (RTL to GDS).

Real-Time FPGA-based Acoustic Imaging*Final Project in Laboratory of Digital Circuit System, GitHub*

Nov., 2022 - Jan., 2023

Taipei, Taiwan

- Implemented a low-complexity delay-and-sum algorithm on Altera DE2-115 FPGA using SystemVerilog to visualize the position of sound sources in real-time via VGA output.
- Demonstrated significant reduction in implementation cost compared to currently high manufacturing costs.

AWARDS AND HONORS**ICCAD Student Scholar Program Grant, ICCAD**

Oct. 2025

Taiwan Chip-based Industrial Innovation Program Research Scholarship, NSTC

Sep. 2024 - Jan. 2025

Scholarship for Overseas Internship Program, NTU / Ministry of Education

June, 2024 - Aug. 2024

Research Scholarship, MediaTek-NTU Research Center

Mar., 2024- June, 2024

Second Prize and Global Nominee, NASA Space App Challenge

Oct., 2023

College Student Research Scholarship, NSTC

Aug., 2023 - Feb., 2024

SKILLS**Languages**

Mandarin (Native), English (Proficient)

Programming Languages

C/C++, Python, Verilog, SystemVerilog, MATLAB, LaTeX, Tcl, Bash

EDA Tools Experience

Synopsys VCS / Design Compiler / PrimeTime, Cadence Innovus